

Step-by-Step Solutions Guide

Exercise 9A (Exercise 9.1)

Chapter 9: Percentage

Based on the curriculum guidelines and exercises from standard mathematics courses

Overview of Exercise 9A: This exercise lays the foundational skills of percentage arithmetic. It covers:

- Converting fractions, decimals, and ratios to percentages.
- Converting percentages back to fractions, decimals, and ratios.
- Calculating specific percentage quantities of given amounts.
- Expressing one quantity as a percentage of another, ensuring appropriate unit conversion beforehand.

Question 1: Converting Fractions into Percentages

Express each of the following as a per cent:

(i) $\frac{3}{5}$

(ii) $\frac{5}{16}$

(iii) $\frac{19}{40}$

(iv) $3\frac{1}{4}$

Solutions:**(i) Express $\frac{3}{5}$ as a per cent:** To convert any fraction to a percentage, multiply it by 100%.

$$\text{Required percentage} = \left(\frac{3}{5} \times 100\right)\% = (3 \times 20)\% = \mathbf{60\%}$$

(ii) Express $\frac{5}{16}$ as a per cent: Multiply by 100%:

$$\text{Required percentage} = \left(\frac{5}{16} \times 100\right)\% = \left(\frac{5 \times 25}{4}\right)\% = \frac{125}{4}\% = 31\frac{1}{4}\% \text{ or } \mathbf{31.25\%}$$

(iii) Express $\frac{19}{40}$ as a per cent: Multiply by 100%:

$$\text{Required percentage} = \left(\frac{19}{40} \times 100\right)\% = \left(\frac{19 \times 5}{2}\right)\% = \frac{95}{2}\% = 47\frac{1}{2}\% \text{ or } \mathbf{47.5\%}$$

(iv) Express $3\frac{1}{4}$ as a per cent: First, convert the mixed fraction to an improper fraction:

$$3\frac{1}{4} = \frac{3 \times 4 + 1}{4} = \frac{13}{4}$$

Multiply by 100%:

$$\text{Required percentage} = \left(\frac{13}{4} \times 100\right)\% = (13 \times 25)\% = \mathbf{325\%}$$

Question 2: Converting Decimals into Percentages

Express each of the following as a per cent:

(i) 0.45

(ii) 0.03

(iii) 0.1

(iv) 2.45

Solutions:**(i) Express 0.45 as a per cent:** To convert a decimal to a percentage, multiply by 100% (which shifts the decimal point two places to the right).

$$\text{Required percentage} = (0.45 \times 100)\% = \mathbf{45\%}$$

(ii) Express 0.03 as a per cent: Multiply by 100%:

$$\text{Required percentage} = (0.03 \times 100)\% = \mathbf{3\%}$$

(iii) Express 0.1 as a per cent: Multiply by 100%:

$$\text{Required percentage} = (0.1 \times 100)\% = \mathbf{10\%}$$

(iv) Express 2.45 as a per cent: Multiply by 100%:

$$\text{Required percentage} = (2.45 \times 100)\% = \mathbf{245\%}$$

Question 3: Converting Ratios into Percentages

Express each of the following as a per cent:

(i) $5 : 8$

(ii) $11 : 18$

(iii) $1 : 5$

(iv) $5 : 4$

Solutions:**(i) Express $5 : 8$ as a per cent:** Write the ratio as a fraction and multiply by 100%:

$$5 : 8 = \frac{5}{8} \implies \left(\frac{5}{8} \times 100 \right) \% = \frac{500}{8} \% = \frac{125}{2} \% = 62\frac{1}{2} \% \text{ or } \mathbf{62.5\%}$$

(ii) Express $11 : 18$ as a per cent: Write as a fraction and multiply by 100%:

$$11 : 18 = \frac{11}{18} \implies \left(\frac{11}{18} \times 100 \right) \% = \frac{1100}{18} \% = \frac{550}{9} \% = \mathbf{61\frac{1}{9} \%} \approx 61.11\%$$

(iii) Express $1 : 5$ as a per cent: Write as a fraction and multiply by 100%:

$$1 : 5 = \frac{1}{5} \implies \left(\frac{1}{5} \times 100 \right) \% = \mathbf{20\%}$$

(iv) Express $5 : 4$ as a per cent: Write as a fraction and multiply by 100%:

$$5 : 4 = \frac{5}{4} \implies \left(\frac{5}{4} \times 100 \right) \% = (5 \times 25) \% = \mathbf{125\%}$$

Question 4: Converting Percentages to Fractions

Express each of the following as a fraction:

(i) 24%

(ii) $12\frac{1}{2}\%$

(iii) $6\frac{2}{3}\%$

(iv) 165%

Solutions:**(i) Express 24% as a fraction:** To convert a percentage to a fraction, divide by 100 and remove the % symbol.

$$24\% = \frac{24}{100} = \frac{24 \div 4}{100 \div 4} = \frac{\mathbf{6}}{\mathbf{25}}$$

(ii) Express $12\frac{1}{2}\%$ as a fraction: Convert mixed fraction: $12\frac{1}{2}\% = \frac{25}{2}\%$.

$$\frac{25}{2}\% = \frac{25}{2 \times 100} = \frac{25}{200} = \frac{25 \div 25}{200 \div 25} = \frac{\mathbf{1}}{\mathbf{8}}$$

(iii) Express $6\frac{2}{3}\%$ as a fraction: Convert mixed fraction: $6\frac{2}{3}\% = \frac{20}{3}\%$.

$$\frac{20}{3}\% = \frac{20}{3 \times 100} = \frac{20}{300} = \frac{20 \div 20}{300 \div 20} = \frac{\mathbf{1}}{\mathbf{15}}$$

(iv) Express 165% as a fraction:

$$165\% = \frac{165}{100} = \frac{165 \div 5}{100 \div 5} = \frac{33}{20} = \mathbf{1\frac{13}{20}}$$

Question 5: Converting Very Small/Decimal Percentages to Fractions

Express each of the following as a fraction:

(i) 0.25%

(ii) 0.05%

(iii) 2.25%

(iv) 0.006%

Solutions:**(i) Express 0.25% as a fraction:**

$$0.25\% = \frac{0.25}{100} = \frac{25}{10000} = \frac{25 \div 25}{10000 \div 25} = \frac{1}{400}$$

(ii) Express 0.05% as a fraction:

$$0.05\% = \frac{0.05}{100} = \frac{5}{10000} = \frac{5 \div 5}{10000 \div 5} = \frac{1}{2000}$$

(iii) Express 2.25% as a fraction:

$$2.25\% = \frac{2.25}{100} = \frac{225}{10000} = \frac{225 \div 25}{10000 \div 25} = \frac{9}{400}$$

(iv) Express 0.006% as a fraction:

$$0.006\% = \frac{0.006}{100} = \frac{6}{100000} = \frac{6 \div 2}{100000 \div 2} = \frac{3}{50000}$$

Question 6: Converting Percentages to Ratios

Express each of the following as a ratio:

(i) 12%

(ii) $\frac{3}{5}\%$

(iii) 7.5%

(iv) 0.36%

Solutions:**(i) Express 12% as a ratio:** First, write as a simplified fraction:

$$12\% = \frac{12}{100} = \frac{3}{25} \implies 3 : 25$$

(ii) Express $\frac{3}{5}\%$ as a ratio:

$$\frac{3}{5}\% = \frac{3}{5 \times 100} = \frac{3}{500} \implies 3 : 500$$

(iii) Express 7.5% as a ratio:

$$7.5\% = \frac{7.5}{100} = \frac{75}{1000} = \frac{75 \div 25}{1000 \div 25} = \frac{3}{40} \implies 3 : 40$$

(iv) Express 0.36% as a ratio:

$$0.36\% = \frac{0.36}{100} = \frac{36}{10000} = \frac{36 \div 4}{10000 \div 4} = \frac{9}{2500} \implies 9 : 2500$$

Question 7: Finding a Certain Percentage of a Given Quantity

Find:

- | | |
|--|--------------------------------|
| (i) 25% of Rs 7 | (ii) $7\frac{1}{2}\%$ of Rs 40 |
| (iii) 1% of Rs 325 | (iv) 8% of 5 kg |
| (v) $16\frac{2}{3}\%$ of 9 m | (vi) 0.6% of 5 km |
| (vii) $6\frac{1}{4}\%$ of (1 hour 20 min.) | |

Solutions:**(i) Find 25% of Rs 7:**

$$25\% \text{ of Rs } 7 = \frac{25}{100} \times 7 = \frac{1}{4} \times 7 = \text{Rs } 1.75 = \text{Rs } 1 \text{ and } 75 \text{ paise}$$

(ii) Find $7\frac{1}{2}\%$ of Rs 40:

$$7\frac{1}{2}\% = \frac{15}{2}\% \implies \frac{15}{2 \times 100} \times 40 = \frac{15}{200} \times 40 = \frac{15 \times 40}{200} = \frac{600}{200} = \text{Rs } 3$$

(iii) Find 1% of Rs 325:

$$1\% \text{ of Rs } 325 = \frac{1}{100} \times 325 = \text{Rs } 3.25 = \text{Rs } 3 \text{ and } 25 \text{ paise}$$

(iv) Find 8% of 5 kg: To express in smaller units, recall 1 kg = 1000 g.

$$8\% \text{ of } 5 \text{ kg} = \frac{8}{100} \times 5 \text{ kg} = 0.4 \text{ kg} = 0.4 \times 1000 \text{ g} = 400 \text{ g}$$

(v) Find $16\frac{2}{3}\%$ of 9 m: Mixed fraction: $16\frac{2}{3}\% = \frac{50}{3}\%$. Recall 1 m = 100 cm.

$$\frac{50}{3 \times 100} \times 9 \text{ m} = \frac{50}{300} \times 9 = \frac{1}{6} \times 9 = 1.5 \text{ m} = 1 \text{ m } 50 \text{ cm}$$

(vi) Find 0.6% of 5 km: Recall 1 km = 1000 m.

$$0.6\% \text{ of } 5 \text{ km} = \frac{0.6}{100} \times 5 \text{ km} = 0.03 \text{ km} = 0.03 \times 1000 \text{ m} = 30 \text{ m}$$

(vii) Find $6\frac{1}{4}\%$ of (1 hour 20 min.): Convert the time to minutes:

$$1 \text{ hour } 20 \text{ min.} = 60 \text{ min.} + 20 \text{ min.} = 80 \text{ minutes}$$

Percentage: $6\frac{1}{4}\% = \frac{25}{4}\%$.

$$\frac{25}{4 \times 100} \times 80 \text{ min} = \frac{25}{400} \times 80 = \frac{1}{16} \times 80 = 5 \text{ minutes}$$

Question 8: Expressing One Quantity as a Percentage of Another

Express:

- (i) 56 p as a per cent of Rs 2.80
- (ii) 45 cm as a per cent of 3 m
- (iii) 630 g as a per cent of 3.5 kg
- (iv) 18 hours as a per cent of 2 days
- (v) 2.8 mm as a per cent of 2.24 cm

Solutions:

(i) Express 56 p as a per cent of Rs 2.80: Convert Rs to paise: Rs 2.80 = $2.80 \times 100 = 280$ paise.

$$\text{Required percentage} = \left(\frac{56}{280} \times 100 \right) \% = \left(\frac{1}{5} \times 100 \right) \% = \mathbf{20\%}$$

(ii) Express 45 cm as a per cent of 3 m: Convert meters to centimeters: 3 m = $3 \times 100 = 300$ cm.

$$\text{Required percentage} = \left(\frac{45}{300} \times 100 \right) \% = \left(\frac{45}{3} \right) \% = \mathbf{15\%}$$

(iii) Express 630 g as a per cent of 3.5 kg: Convert kg to grams: 3.5 kg = $3.5 \times 1000 = 3500$ grams.

$$\text{Required percentage} = \left(\frac{630}{3500} \times 100 \right) \% = \left(\frac{630}{35} \right) \% = \frac{630 \div 7}{35 \div 7} \% = \frac{90}{5} \% = \mathbf{18\%}$$

(iv) Express 18 hours as a per cent of 2 days: Convert days to hours: 2 days = $2 \times 24 = 48$ hours.

$$\text{Required percentage} = \left(\frac{18}{48} \times 100 \right) \% = \left(\frac{3}{8} \times 100 \right) \% = \frac{300}{8} \% = 37\frac{1}{2} \% \text{ or } \mathbf{37.5\%}$$

(v) Express 2.8 mm as a per cent of 2.24 cm: Convert cm to mm (1 cm = 10 mm): 2.24 cm = $2.24 \times 10 = 22.4$ mm.

$$\text{Required percentage} = \left(\frac{2.8}{22.4} \times 100 \right) \% = \left(\frac{28}{224} \times 100 \right) \% = \left(\frac{1}{8} \times 100 \right) \% = \mathbf{12.5\%} \text{ or } 12\frac{1}{2} \%$$

Question 9: Advanced Quantity Percentage Comparisons

Answer each of the following:

- (i) What per cent of 1.5 m is 60 cm?
- (ii) What per cent of 2 kg is 125 g?
- (iii) What per cent of Rs 6 is 40 p?
- (iv) What per cent of a day is 10 h?
- (v) What per cent of $7\frac{1}{3}$ m is $5\frac{1}{2}$ m?

Solutions:

(i) What per cent of 1.5 m is 60 cm? Convert 1.5 m to cm: $1.5 \times 100 = 150$ cm.

$$\text{Required percentage} = \left(\frac{60}{150} \times 100 \right) \% = (0.4 \times 100)\% = \mathbf{40\%}$$

(ii) What per cent of 2 kg is 125 g? Convert 2 kg to grams: $2 \times 1000 = 2000$ g.

$$\text{Required percentage} = \left(\frac{125}{2000} \times 100 \right) \% = \frac{125}{20} \% = 6\frac{5}{20} \% = \mathbf{6\frac{1}{4}\% \text{ or } 6.25\%}$$

(iii) What per cent of Rs 6 is 40 p? Convert Rs 6 to paise: $6 \times 100 = 600$ paise.

$$\text{Required percentage} = \left(\frac{40}{600} \times 100 \right) \% = \frac{40}{6} \% = \frac{20}{3} \% = \mathbf{6\frac{2}{3}\% \approx 6.67\%}$$

(iv) What per cent of a day is 10 h? Convert 1 day to hours: 24 hours.

$$\text{Required percentage} = \left(\frac{10}{24} \times 100 \right) \% = \left(\frac{5}{12} \times 100 \right) \% = \frac{500}{12} \% = \frac{125}{3} \% = \mathbf{41\frac{2}{3}\% \approx 41.67\%}$$

(v) What per cent of $7\frac{1}{3}$ m is $5\frac{1}{2}$ m? First, convert the mixed numbers to improper fractions:

$$\text{First Quantity} = 7\frac{1}{3} \text{ m} = \frac{22}{3} \text{ m}$$

$$\text{Second Quantity (to be compared)} = 5\frac{1}{2} \text{ m} = \frac{11}{2} \text{ m}$$

The percentage formula is $\frac{\text{Second Quantity}}{\text{First Quantity}} \times 100\%$:

$$\text{Required percentage} = \left(\frac{\frac{11}{2}}{\frac{22}{3}} \times 100 \right) \% = \left(\frac{11}{2} \times \frac{3}{22} \times 100 \right) \%$$

Simplify:

$$= \left(\frac{11}{22} \times \frac{3}{2} \times 100 \right) \% = \left(\frac{1}{2} \times \frac{3}{2} \times 100 \right) \% = \left(\frac{3}{4} \times 100 \right) \% = 3 \times 25 = \mathbf{75\%}$$